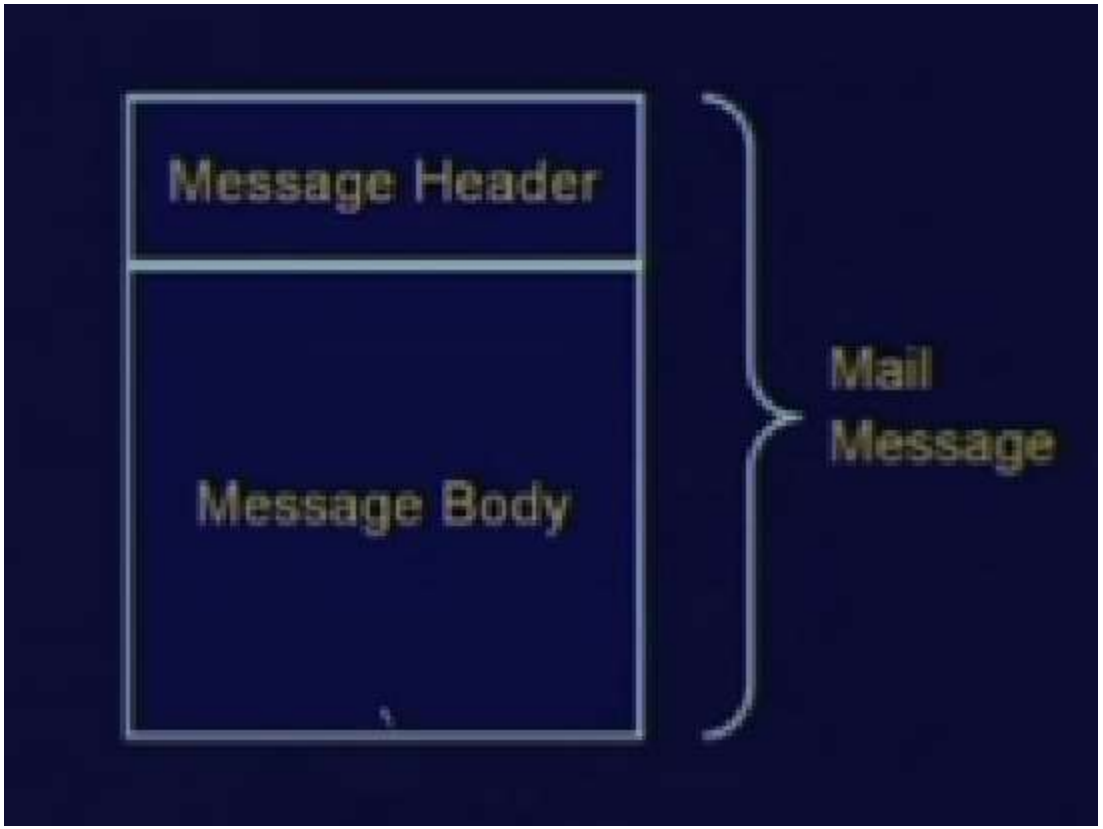
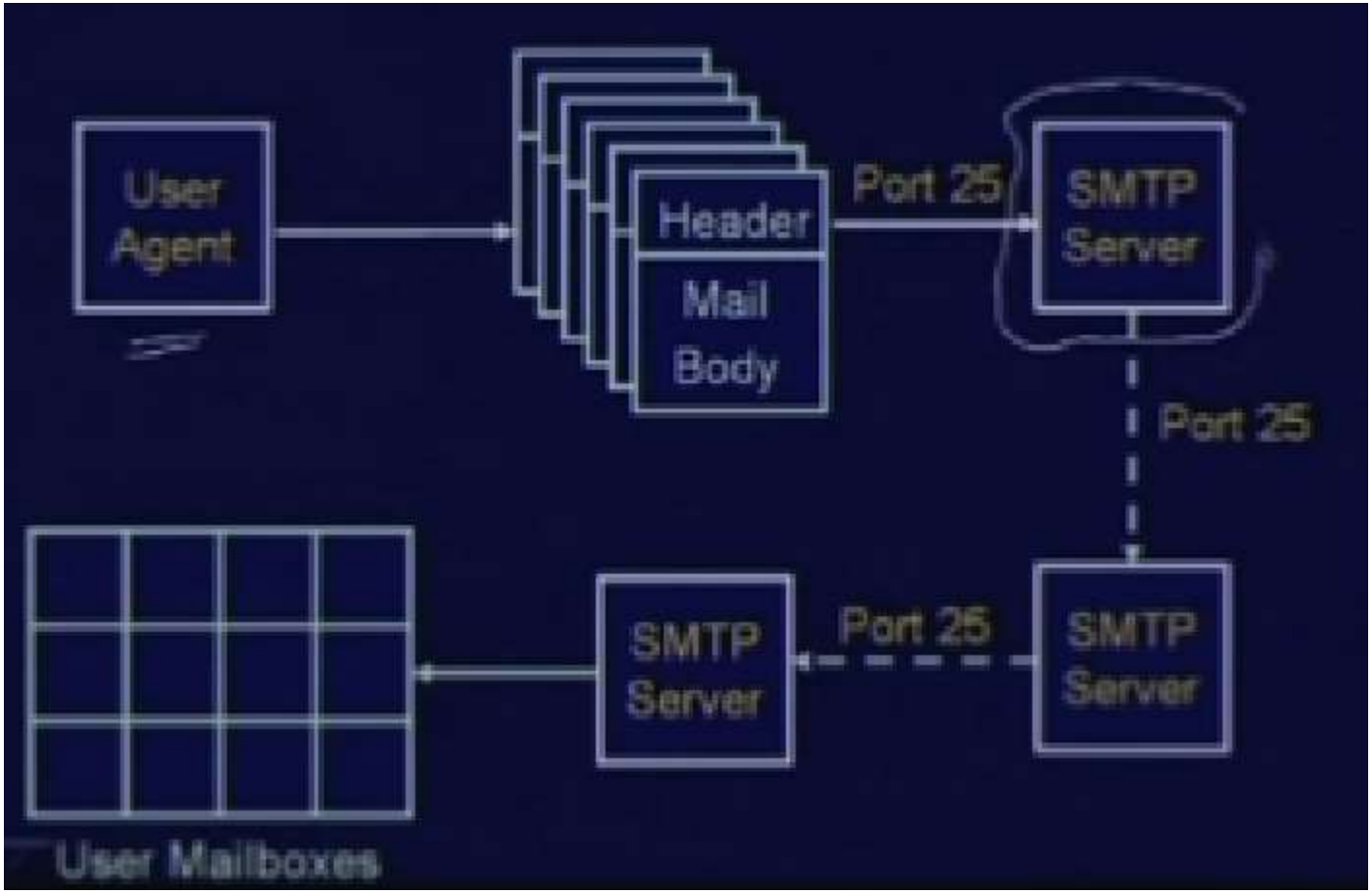
Simple Mail Transfer Protocol

- Based on RFC 821.
- Transmits simple text messages only.
  - 7-bit ASCII format.
- Uses information written on envelope of mail.
  - Message header.
  - Contains recipient address and other information.
- Does not look at contents.
  - Message body.


Message Header
Message Body
Mail
Message

Basic Operation

- Mail is created by user agent program (mail client).
- Messages queued and sent as input to SMTP sender program.
  - Typically a server process.
  - Daemon on UNIX.
    - sendmail or qmail


User Agent
Header
Mail Body
Port 25
SMTP Server
Port 25
SMTP Server
Port 25
SMTP Server
User Mailboxes

Mail Message Contents

- Each queued message has:
  - Message text
    - RFC 822 header with message envelope and list of recipients.
    - Message body, composed by user.
  - A list of mail destinations
    - Derived by user agent / SMTP server from header.
    - May require expansion of mailing lists.

SMTP Sender

- Takes message from queue.
- Transmits to proper destination host.
  - Via SMTP transaction.
  - Over one or more TCP connections to port 25.
- When all destinations processed, message is deleted.

Optimization

- If message is sent to multiple users on a given host, it is sent only once.
  - Delivery to users handled at destination host.
- If multiple messages are ready for given host, a single TCP connection can be used.
  - Saves overhead of setting up and dropping connection.

Possible Errors

- Host unreachable
- Host out of operation
- TCP connection fail during transfer
- Faulty destination address
  - User error
  - Target user address has changed
  - Redirect if possible
  - Inform user if not

Sender can re-queue mail

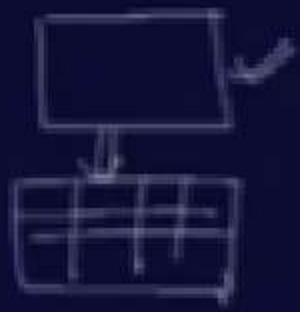
- Give up after a period

SMTP Protocol - Reliability

- Used to transfer messages from sender to receiver over TCP connection.
  - Uses port number 25.
- Attempts to provide reliable service.
- No guarantee to recover lost messages.
- No end-to-end ACK to sender.
- Error indication report not guaranteed.

SMTP Receiver

- Accepts arriving message.
- Places in user mailbox or copies to outgoing queue for forwarding.
- Receiver must:
  - Verify local mail destinations.
  - Deal with errors
    - Transmission ,
    - Lack of disk space



SMTP Forwarding

- Mostly direct transfer from sender host to receiver host.
- May go through intermediate mail servers via forwarding capability.
  - Sender can specify route.

SMTP System Overview

- Commands and responses exchanged between sender and receiver.
- Initiative with sender.
  - Establishes TCP connection.
- Sender sends commands to receiver.
  - e.g. HELO <domain><CRLF>
- Each command generates exactly one reply.
  - e.g. 250 requested mail action ok; completed.

SMTP Replies

- Starts with 3-digit code.
- Leading digit indicates category.
  - 2xx -- Positive completion reply
  - 3xx -- Positive intermediate reply
  - 4xx -- Transient negative completion reply
  - 5xx -- Permanent negative completion reply

Operation Phases

a) Connection setup
b) Exchange of command-response pairs
c) Connection termination

a) Connection Setup

- Sender opens TCP connection with receiver. telnet 2.
- Once connected, receiver identifies itself.

  220 <domain> service ready
- Sender identifies itself.

  HELO
- Receiver accepts sender's identification.

  250 OK
- If mail service not available, the second step above becomes:

  421 service not available

b) Mail Transfer Commands

- The MAIL FROM command identifies originator.
  - Gives reverse path to be used for error reporting.
  - Receiver returns 250 OK or appropriate failure / error message.

c) Closing Connection

- Two steps:
  - Sender sends QUIT and waits for reply.
  - Then initiate TCP close operation.
- Receiver initiates TCP close after sending reply to QUIT.

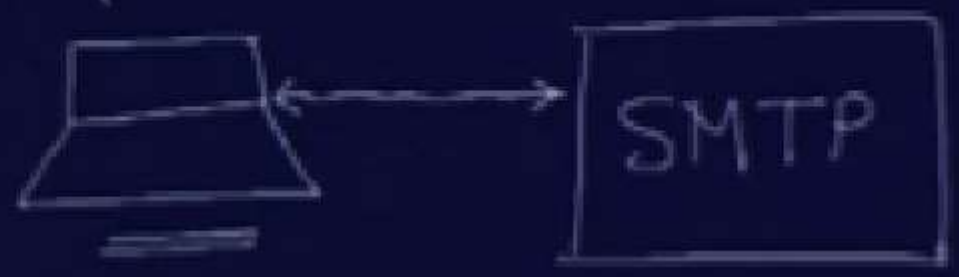
- How to connect to an SMTP server?

  telnet servername 25

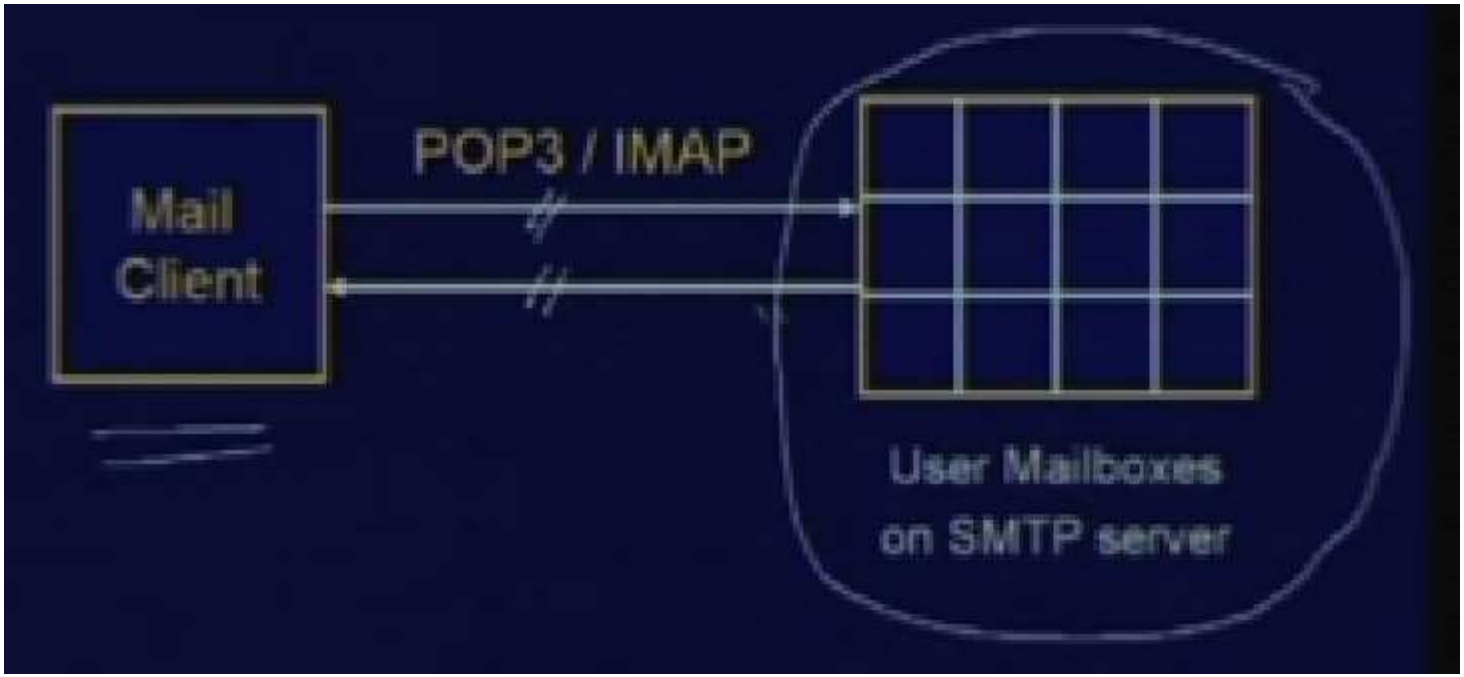
  - A TCP connection gets established over port number 25.
  - The telnet client and the mail server can now start a dialogue.

Mail Access Protocols

- Two mail access protocols are widely used:
  1. Post Office Protocol, version 3 (POP3)
  2. Internet Mail Access Protocol version 4 (IMAP4).



What do they do?



POP3

- The client POP3 software is installed on the recipient machine, and the server POP3 software installed on mail server.
  - The client (user agent) opens a connection with the server on TCP port number 110.
  - Sends user name and password.
  - Can access the mails, one by one.

POP3 (contd.)

- Two modes:
  - Delete mode – mails deleted as they are read
  - Keep mode – mails remain in the mailbox
- POP3 has commands for:
  - Log in
  - Log out
  - Fetch messages
  - Delete messages

IMAP4

- Provides the following extra features:
  - A user can check the email header before downloading.
  - A user can search the contents of the email for a specific string prior to downloading.
  - A user can create, delete, or rename mailboxes on the mail server.
  - A user can create a hierarchy of mailboxes in a folder for email storage.

Multipurpose Internet Mail Extension (MIME)

- SMTP cannot transmit non-text messages.
  - Solutions (like uuencode) exists on some systems, but are not standardized.
- Cannot transmit text that includes international characters (e.g. â, á, ä, è, é, ê, ë).
  - Need 8 bit ASCII.

Overview of MIME

- Five new message header fields:
  - MIME-version
  - Content-type
  - Content-transfer-encoding
  - Content-Id
  - Content-description
- A number of *content types* and *transfer encoding* formats have been defined.

Content Types

- Text body
- Multipart
  - Mixed, Parallel, Alternative
- Message
  - RFC 822, Partial, External-body
- Image
  - jpeg, gif
- Video
  - mpeg
- Audio
  - Basic
- Application
  - Postscript
  - octet stream

MIME Transfer Encodings

- Specifies how the mail body is wrapped for transmission.
- Content transfer encoding field can have six possible values.
  - 7bit, 8bit, binary: no encoding done for these three.
    - Provide information about nature of data.

What is a domain?

A set of computers related by either their geographic location, or the way they function.

What are named and nslookup?

named: domain name server running under the UNIX system.

nslookup: command to invoke DNS name resolver.